# Exact diffusion scores for non-Gaussian Markov chains: what second-order message closure computes

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## Abstract

The global minimiser of the objective that trains a diffusion model is the score of the *empirical* distribution, not of the data distribution — a model attaining it would reproduce its training set and nothing else. Deciding what a network actually learned means comparing it against the score it should have learned, which for realistic data is unavailable. We exhibit a family where it is available and non-trivial: a first-order Markov chain with continuous, non-Gaussian transitions. Coordinatewise noising multiplies the prior by *unary* factors, so the posterior factor graph is still the prior's chain — a tree, on which sum-product is exact. The exact score is therefore available for a correlated, non-Gaussian model, and we validate the grid representation of the continuous messages against the Gaussian closed form and against brute-force enumeration of the discretised posterior. We then prove that single-Gaussian message closure computes *exactly* the LMMSE estimator of the covariance-matched Gaussian model, so the gap between it and full sum-product is precisely the price of discarding everything beyond second moments — not an uncontrolled approximation but a named estimator. The same posterior statistics also supply the exact likelihood gradient needed to learn the transition, which the longer paper develops.

## 1 The problem

Diffusion models are trained by regressing the posterior mean of a clean signal given its noised version [1–4]. A network never sees the data distribution $P_0$; it sees $M$ samples, and the global minimiser of that regression is the score of the empirical measure $\hat{P}_0 = M^{-1} \sum_\mu \delta(\cdot - \mathbf{a}^\mu)$ smoothed by the forward Gaussian. A model attaining it exactly would generate its training set and nothing else. That diffusion models instead generate new data is the memorisation–generalisation question studied in the statistical mechanics of these models [5, 6].

Progress on it needs a target one can compute. For realistic data the exact score is unavailable, so a network's error cannot be separated from the difficulty of what it was approximating. Tractable families are scarce and mostly Gaussian, where the score is linear and every question about structure has a degenerate answer.

**This work.** We use a first-order Markov chain: diffusion is applied to sequential data, and studying how structure is exploited requires data that has structure one can dial. The chain is variance-matched, so Gaussian, Laplace, Student, uniform and bimodal innovations share a covariance exactly and differ only beyond second moments. Two things follow, and this note is about the first: the exact score is computable by local inference, and the standard second-order approximation to that inference turns out to be an identifiable estimator rather than an uncontrolled one. The same machinery also learns the kernel when it is unknown, which the longer paper develops.

## 2 The model, and why its score is computable

**Forward process.** With $\eta$ standardised Gaussian white noise, $\langle \eta(t)\eta(t')\rangle = \delta(t-t')$, the forward Ornstein–Uhlenbeck process and its solution are

$$\frac{\mathrm{d}\mathbf{x}}{\mathrm{d}t} \;=\; -\mathbf{x} + \sqrt{2}\,\eta(t), \qquad \mathbf{x}(t) \;=\; e^{-t}\mathbf{a} + \sqrt{\Delta_t}\,\mathbf{z}, \qquad \Delta_t = 1 - e^{-2t}, \quad \mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})\,. \tag{1}$$

The reverse-time convention is fixed once in Appendix A and used without variation.

**Prior.** $P_0$ factorises as

$$P_0(\mathbf{a}) \;=\; \mu(a_1)\prod_{i=2}^{L} K(a_i \mid a_{i-1}), \qquad K(a' \mid a) \;=\; \varphi(a' - \rho a), \tag{2}$$

with $\varphi$ an innovation density of mean zero and variance $q$, and $|\rho| < 1$. We draw $a_1 \sim \mathcal{N}(0,1)$ and set $q = 1 - \rho^2$, which makes $\mathrm{Var}(a_i) = 1$ at every site and $\mathrm{Cov}(a_i, a_j) = \rho^{|i-j|}$ exactly, *whatever the shape of* $\varphi$. Variance matching is what makes the family a controlled probe: Gaussian, Laplace, Student, uniform and bimodal innovations share the same covariance and differ only beyond second moments. The chain is covariance-stationary but not strictly stationary; Appendix C states the difference, and shows why it collapses in the Gaussian case.

**The score is a posterior mean.** From (1), $P_t$ is a Gaussian convolution of $P_0$. Differentiating under the integral and dividing by $P_t(\mathbf{x})$ — the Gaussian factor over $P_t(\mathbf{x})$ is exactly the posterior density $P(\mathbf{a} \mid \mathbf{x}, t)$ — gives

$$S(\mathbf{x}, t) \;=\; -\,\frac{\mathbf{x} - e^{-t}\,\langle \mathbf{a}\rangle_{x,t}}{\Delta_t}, \qquad \langle \mathbf{a}\rangle_{x,t} = \int \mathrm{d}\mathbf{a}\;\mathbf{a}\,P(\mathbf{a} \mid \mathbf{x}, t). \tag{3}$$

Equation (3) is a change of variables, not an approximation: computing the score and computing the posterior mean are the same problem. The identity is classical in the empirical-Bayes literature [7–9] and underlies denoising score matching [4, 10]. For squared loss $\langle \mathbf{a}\rangle_{x,t}$ minimises the conditional risk (Appendix B), so it is the reference every estimator below is scored against.

The likelihood factors are *unary*: the forward noise is independent across coordinates, so each observation touches one latent variable. Unary factors reweight node potentials without creating edges among the $a_i$, so conditioning leaves the factor graph a chain — a tree, on which sum-product returns exact marginals [11–13]. On a chain these are the forward–backward recursions [14, 15], reducing to the Kalman filter when everything is Gaussian [16, 17].

Exactness is at the level of *functional* messages. A message is a function on $\mathbb{R}$, forced by the state space, so an implementation must represent one; we use a truncated grid of $N_\mathrm{g}$ points, which makes the recursion exact for a finite-state model whose discretisation error is measured separately. At $N_\mathrm{g} = 401$ it reproduces the Gaussian closed form to $9.2 \times 10^{-15}$ and brute-force enumeration of the discretised posterior to $1.6 \times 10^{-14}$. Cost is $O(LN_\mathrm{g}^2)$ per sequence against $O(N_\mathrm{g}^{L-1})$ for direct summation.

**What the standard Gaussian approximation computes.** Propagating only the first two moments of each message is the usual approximation. On this model it has an exact characterisation, which turns a heuristic into a measurement:

**Proposition 1.** *For the chain (2) with linear transitions, Gaussian unary likelihoods and* $\mathbb{E}[\mathbf{a}] = \mathbf{0}$*, moment-projected sum-product returns the LMMSE estimator of the covariance-matched Gaussian model,*

$$\widehat{\mathbf{a}}_{\mathrm{LMMSE}} = \mathrm{Cov}(\mathbf{a}, \mathbf{x})\;\mathrm{Cov}(\mathbf{x})^{-1}\mathbf{x} = e^{-t}\boldsymbol{\Sigma}_0\left(e^{-2t}\boldsymbol{\Sigma}_0 + \Delta_t\mathbf{I}\right)^{-1}\mathbf{x}, \tag{4}$$

*for every innovation law with mean zero and variance* $q$.

So the gap between moment-projected and full sum-product is precisely the price of replacing the prior by its covariance twin — not an artefact of a message representation. It vanishes as $t$ grows, because noising Gaussianises the posterior, so the shape of $\varphi$ survives only where the likelihood is weak: on the Laplace chain the median relative score error falls from $0.241$ at $t = 0.05$ to $7.3 \times 10^{-6}$ at $t = 3$ (Figure 1).

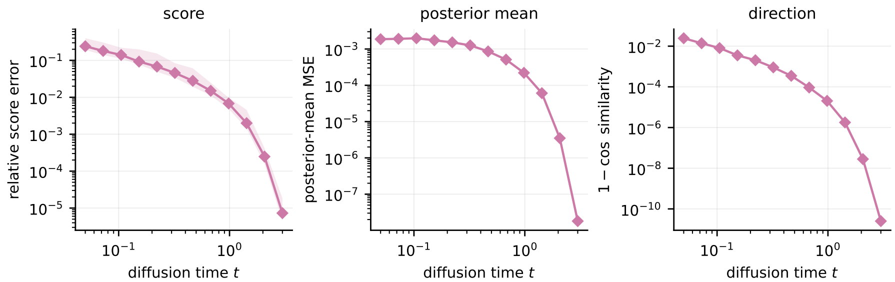


Figure 1: What the Gaussian second-order baseline costs, against sum-product under the true kernel. By Proposition 1 this gap is exactly the price of replacing the prior by its covariance-matched Gaussian, so it measures non-Gaussianity rather than approximation quality. Laplace chain, $\rho = 0.85$; median over seeds with the 10–90 percentile band.

## 3 The same inference also learns the chain

Everything above assumes the kernel is known. It need not be. The pairwise posterior marginals that sum-product already produces are exactly the expected transition statistics that likelihood learning needs: writing $\mathcal{L}(\theta) = \sum_{\mu} \log P_{t,\theta}(\mathbf{x}^{\mu})$, Fisher's identity gives $\nabla_{\theta}\mathcal{L}$ as a posterior expectation of the complete-data score, so one forward–backward pass yields the exact gradient and no differentiation through the $2(L-1)$ message updates is required. Accumulated over edges the pairwise beliefs form a single matrix $\boldsymbol{\Xi}$ that is a complete summary for the resulting surrogate objective, so the maximisation step never revisits the data; for Gaussian innovations it is closed form, the belief-weighted Yule–Walker equations. The derivations, the mixture-innovation extension, and the recovery experiments are in the longer paper.

## 4 Scope, and what is open

Exactness is structural, not numerical: the recursion is exact on the functional messages, and what an implementation actually runs is a truncated grid whose truncation and quadrature error must be measured rather than assumed. The figures quoted above are agreements at the stated configuration, not a bound for arbitrary kernels, tails or noise levels — trapezoidal quadrature is second order, and it degrades when a likelihood is narrower than a few grid cells. Everything here also assumes exact first-order Markov structure; under a non-Markov prior the inference is misspecified in a way more data does not repair. Inference costs $O(LN_{\mathrm{g}}^2)$ per sequence, well above a network forward pass, so what this model buys is a computable target, not speed.

The open question is architectural. Sum-product on a chain suggests a network narrow but as deep as twice the chain, unlike the tree-structured case where the natural depth is the tree's [18, 19]. Which architectures exploit Markovianity, and at what depth, is what this setting is built to measure.

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# A Notation

$L$ is the chain length, which is also the ambient dimension; $M$ is the number of observed sequences; $N_{\mathrm{g}}$ is the number of quadrature points; $K$ is always the Markov transition kernel, $\varphi$ the innovation density, $\rho$ the autoregressive coefficient and $q$ its variance. Vectors are bold lowercase, matrices bold uppercase. The forward process is $\mathbf{x}(t) = e^{-t}\mathbf{a} + \sqrt{\Delta_t}\,\mathbf{z}$ with $\Delta_t = 1 - e^{-2t}$; the reverse SDE and its sign conventions follow Anderson [20], Haussmann and Pardoux [21] and are stated in full in the companion paper.

# B The score–posterior identity

$P_t$ is a Gaussian convolution of $P_0$, and $\nabla_{\mathbf{x}} g_{\Delta_t}(\mathbf{x} - e^{-t}\mathbf{a}) = -\frac{\mathbf{x}-e^{-t}\mathbf{a}}{\Delta_t} g_{\Delta_t}(\mathbf{x} - e^{-t}\mathbf{a})$, with the integrand dominated uniformly whenever $P_0$ has a finite first moment. Dominated convergence then licenses differentiation under the integral; dividing by $P_t(\mathbf{x})$ and recognising the posterior density gives (3). This is Tweedie's formula [7–9]. Since $\langle \mathbf{a} \rangle_{x,t}$ minimises the conditional squared risk, every error below is reported *against* it rather than as raw risk: the Bayes floor is common to all estimators, and subtracting it is what makes the comparison informative.

# C Stationarity

With $q = 1 - \rho^2$ and $\mathrm{Var}(a_1) = 1$, second moments propagate exactly: $\mathrm{Var}(a_i) = 1$ and $\mathrm{Cov}(a_i, a_j) = \rho^{|i-j|}$ for every innovation shape. The chain is therefore *covariance*-stationary, which is what makes the variance-matched family a controlled probe. It is not strictly stationary for non-Gaussian $\varphi$ — the invariant law of $a_i = \rho a_{i-1} + \varepsilon_i$ is that of $\sum_{k \geq 0} \rho^k \varepsilon_{-k}$, which is not Gaussian — and we do not claim it is. The two notions coincide for Gaussians because a Gaussian law is determined by its first two moments.

# D Protocol

One frozen configuration, defined in a single file and imported by every experiment so that no run can diverge from it: Laplace chain unless stated, $\rho = 0.85$, $L = 32$, $N_{\mathrm{g}} = 401$, $A = 8$, twelve log-spaced noise levels on $[0.05, 3.0]$, budgets $M \in \{32, \ldots, 4096\}$, and 256 held-out sequences on a disjoint seed. The ring uses $T = 12$. EM stops when the fitted innovation excess kurtosis stops moving, $|\Delta\kappa_4| < 10^{-3}$, never when $\rho$ does: a settled $\rho$ certifies a converged-looking trace over a kernel that is nowhere near converged.